

Mohnish Shende

Cybersecurity Engineer | Security Researcher

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Professional Summary

Early-career cybersecurity engineer and security researcher focused on vulnerability assessment, infrastructure hardening, OSINT, and security tooling. Builds reproducible self-hosted infrastructure and offline security tools, with peer-reviewed research connecting cybersecurity, forensic psychology, and behavioral threat assessment. Experience includes vulnerability assessment, technical reconnaissance, responsible disclosure to CERT-In, and behavioral threat assessment research.

Professional Experience

Project Associate I

CSIR National Environmental Engineering Research Institute

Dec 2024 - Mar 2025 | Nagpur, India

- Performed vulnerability assessments using Nessus and Burp Suite.
- Supported network segmentation and access-control policy implementation.
- Documented secure configuration baselines and system hardening guidelines.
- Prepared technical reports for security assessments and infrastructure hardening work.

Technical Skills

Security Engineering: VAPT, vulnerability assessment, incident response, digital forensics, OSINT, network security, technical reconnaissance, infrastructure hardening, responsible vulnerability disclosure

Infrastructure: Debian 13, Linux, Windows Server, networking, VPNs, firewalls, reverse proxies, Docker, Docker Compose, self-hosted infrastructure, persistent storage separation

Security Tools: Nmap, Nessus, Burp Suite, Wireshark, Metasploit, Pi-hole, Unbound, Caddy, Tailscale, Ollama, Git

Standards & Methodologies: MITRE ATT&CK, OWASP Top 10, NIST SP 800-61, CVSS, ISO 27001

Programming: Python, Bash, SQL, Flask, XML, JSON, Markdown

Research: Threat intelligence, behavioral threat assessment, forensic psychology, cyber psychology

Security Engineering Projects

NodeZero Homelab Infrastructure Platform

github.com/MohnishShende/NodeZero

Self-hosted infrastructure platform | Debian 13, Docker Compose, Caddy, Pi-hole, Unbound, Tailscale

- Designed and deployed a containerized Debian 13 platform using Docker Compose with isolated service boundaries.
- Configured recursive DNS and DNS-based filtering with Pi-hole and Unbound, using Caddy for reverse proxy and HTTPS routing.
- Established encrypted remote access with Tailscale and documented disaster recovery and rebuild workflows.

IBI Schemes

github.com/MohnishShende/IBI-schemes

Applied cryptography | Python 3.10+, Elliptic Curve Cryptography, Flask, cryptography, tinyec

- Implemented five Identity-Based Identification (IBI) protocols in Python, including Twin Schnorr, Modified Schnorr, Sakai Kasahara, Fiat Shamir, and Zero-Knowledge schemes.
- Profiled runtime and memory usage (`tracemalloc`) for key generation, proving, and verification across pairing-based and pairing-free implementations.
- Built a Flask client-server simulation to model identity authentication workflows.

Nmap LLM

github.com/MohnishShende/nmap-llm

AI security tooling | Python 3.10+, Nmap, Ollama, Mistral, XML, JSON, Markdown

- Built an offline, local LLM-assisted Nmap workflow (Ollama/Mistral) for scan planning, command generation, and results triage.
- Built a Python CLI that converts natural-language inputs into validated Nmap commands, enforcing strict private IPv4 range boundaries.
- Automated reporting that parses Nmap XML to JSON and generates Markdown triage reports with timestamped execution records.

Peer-Reviewed Publication

Psychological Appropriation: A Dimensional Model of Surrogate-Moral Violence

Peer-reviewed journal article, Journal of Forensic Psychology Research and Practice, Taylor & Francis / Informa UK Limited, 2025

doi.org/10.1080/24732850.2025.2599908

- Formulated a conceptual framework analyzing surrogate-moral violence and psychological appropriation.
- Proposed analytical dimensions for behavioral threat assessment.
- Synthesized perspectives from forensic psychology, criminology, behavioral analysis, and threat assessment.

Security Research and Responsible Disclosure

- Performed independent OSINT and public attack-surface assessments, analyzing infrastructure metadata and exposed services.
- Authored and submitted a structured responsible disclosure dossier to CERT-In documenting multiple security observations identified through passive reconnaissance of a public-sector web application, with technical evidence, risk analysis, and remediation recommendations.
- Received follow-up from CERT-In requesting additional validation material.
- Engineered a [version-controlled forensic-intelligence research repository](#) integrating LaTeX publishing, source and evidence registers, claim-verification controls, chronology reconstruction, structured analysis, and release governance.
- Researched behavioral analysis frameworks for politically motivated cyber and physical threats.

Education

MSc Cyber Security, Merit

University of Plymouth

NCSC Certified Master's Degree

Awarded 2024 | United Kingdom

BE Information Technology, Distinction

Rashtrasant Tukadoji Maharaj Nagpur University

Completed 2022 | Nagpur, India